

Denmark Bird Biodiversity Hotspot Monitoring Dashboard

Mini Project for DS3294: Data Science Practice

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1 Introduction

Biodiversity monitoring at national scales has historically relied on structured surveys conducted by trained ecologists, a process that is expensive, slow, and geographically sparse. Over the past two decades, citizen-science platforms have dramatically expanded the spatial and temporal coverage of biodiversity records by engaging amateur naturalists as data collectors [1]. Denmark's DOFbasen database, administered by Dansk Ornitologisk Forening (DOF/BirdLife Denmark), exemplifies this paradigm: it contains tens of millions of bird observation records spanning several decades, contributed by thousands of volunteer observers across the country.

However, the scientific value of citizen-science data is inseparable from its limitations. Records are opportunistically collected — observers go where they choose, when they choose — producing a dataset whose spatial and temporal coverage reflects human behaviour as much as ecological reality. Regions near cities and popular birdwatching routes accumulate disproportionately more records than remote or ecologically similar but less accessible areas. This *observer effort bias* means that raw species counts per area cannot be interpreted as unbiased estimates of true biodiversity without appropriate correction.

In this project, we address these challenges through a complete data science pipeline applied to the DOFbasen dataset. We (1) ingest the data efficiently via a memory-constrained streaming pipeline, (2) clean and standardise records addressing four categories of data quality problems, (3) define two parallel spatial aggregation frameworks for Denmark, (4) Build an interactive dashboard that makes these results explorable.

The remainder of this report is structured as follows. Section 2 reviews related work on citizen-science data analysis and biodiversity hotspot identification. Section 3 summarises our specific contributions. Section 4 describes our data and methods in detail. Section 5 presents our results, and Section 8 concludes with a discussion of limitations and future directions.

2 Related Work

Bird biodiversity patterns have been widely studied using spatial datasets derived from observational records. Large-scale bird monitoring initiatives and citizen science platforms have produced extensive datasets that allow researchers to analyze spatial distributions of bird species.

Previous studies have shown that species richness patterns are influenced by factors such as habitat diversity, climate conditions, and land-use patterns. In northern European countries such as Denmark, coastal regions, wetlands, and migratory stopover sites often support high bird diversity.

Spatial grid-based approaches are commonly used in ecological studies to map species richness and identify biodiversity hotspots. By aggregating species observations within geographic

grid cells, researchers can visualize biodiversity patterns and identify regions that may require conservation attention.

3 Our Contribution

The primary contributions of this project are as follows.

Memory-efficient streaming ingestion pipeline. We develop a Python pipeline that streams the 35-million-record DOFbasen dataset line by line from a remote HuggingFace-hosted file, applies quality filters during reading, and writes accepted records to disk in batches using PyArrow’s `ParquetWriter`. This approach maintains a flat RAM profile regardless of dataset size, making the pipeline reproducible on standard laptops without high-performance computing resources.

Systematic data quality assessment. We characterise and address four categories of data quality problems specific to citizen-science occurrence data: taxonomic inconsistencies, spatial coordinate uncertainty, duplicate records, and missing metadata. Each cleaning decision is documented with an explicit justification and quantified in terms of records removed.

Dual spatial aggregation framework. We implement and maintain two parallel spatial representations of the data throughout the analysis pipeline: a regular 0.3° grid and Denmark’s five administrative regions.

Biodiversity indicators. We compute species richness, range restriction (occupancy), and observation effort under both spatial frameworks.

National park overlap analysis. We perform a geometric intersection analysis between hotspot grid cells and Denmark’s five national park polygons (obtained from OpenStreetMap via Overpass Turbo), identifying the extent to which existing protected areas cover observed biodiversity peaks and where gaps exist.

Interactive dashboard. We deliver an interactive Dash application that allows users to explore species distributions, richness maps, effort maps, and hotspot designations through a browser-based interface backed by pre-computed summary files.

4 Methods

4.1 Data

The primary dataset is DOFbasen, an online citizen-science database of biodiversity observations in Denmark administered by Dansk Ornitologisk Forening (DOF/BirdLife Denmark) [2]. The dataset is distributed via the Global Biodiversity Information Facility (GBIF) and contains approximately 35 million records covering birds, mammals, amphibians, reptiles, and butterflies collected by volunteer observers over several decades. Records are casual observations, meaning observers did not follow a standardised protocol and did not necessarily report every species or individual encountered. Rare and unusual bird observations are subjected to quality assessment by two Rarity Committees; dismissed or pending observations are excluded from the published dataset.

The dataset was accessed via a HuggingFace-hosted mirror at the URL https://huggingface.co/datasets/itsmeG/Birds_denmark/resolve/main/Database/occurrence.txt and is distributed in Darwin Core Archive format, with records stored as a tab-separated values file with 50+ columns following the Darwin Core standard vocabulary.

4.2 Data Ingestion

Given the dataset’s size (approximately 35 million rows), direct in-memory loading is not feasible on standard hardware. We implement a streaming pipeline using the Python `requests` library

Filter	Criterion
Taxonomic class	<code>class == "Aves"</code>
Coordinate validity	<code>hasCoordinate == "true"</code>
Geospatial issues	<code>hasGeospatialIssues == "false"</code>
Occurrence status	<code>occurrenceStatus == "PRESENT"</code>
Taxonomic rank	<code>taxonRank == "SPECIES"</code>
Species name	Non-empty <code>species</code> field
Year	<code>year >= 2000</code>

Table 1: Inline quality filters applied during streaming ingestion. All values are compared as strings because values from the tab-separated stream are raw strings prior to type conversion.

with `stream=True`, which reads the file one line at a time over an HTTP connection without downloading the entire file to disk. Column indices are extracted from the header row and used to parse only the 20 columns required for analysis, avoiding the memory cost of loading unused fields.

Quality filters are applied *during* streaming, meaning rejected rows are never accumulated in memory. Accepted rows are buffered in a list of dictionaries and flushed to disk in batches of 100,000 rows using `pyarrow.parquet.ParquetWriter`, which appends each batch to a single Parquet output file. This architecture maintains peak RAM usage below approximately 100MB regardless of the total dataset size. The filters applied during ingestion are summarised in Table 1.

4.3 Data Cleaning

Following ingestion, we apply four categories of data cleaning to the filtered dataset.

Taxonomic inconsistencies. Records flagged with `TAXON_MATCH_FUZZY` in the `GBIF issue` column indicate that GBIF’s automated pipeline could not match the submitted species name exactly to its taxonomic backbone and made a best-guess assignment. Such records carry taxonomic identification uncertainty that makes them unreliable for species-level analysis. These records are excluded. Additionally, the `taxonRank == SPECIES` filter applied at ingestion preemptively excludes all records identified only to genus, family, or higher taxonomic levels.

Spatial uncertainty. The `coordinateUncertaintyInMeters` column quantifies GPS coordinate imprecision. Records with uncertainty exceeding 10,000 metres are excluded, as at our 0.3° grid resolution (≈ 25 km per cell in Denmark), records with > 10 km uncertainty could be assigned to an incorrect grid cell. We additionally verify that all retained coordinates fall within Denmark’s geographic bounding box (latitude 54.5–57.8°N, longitude 8.0–15.3°E).

Duplicate records. Duplicate records are identified as rows sharing identical values in the combination of `species`, `eventDate`, `decimalLatitude`, and `decimalLongitude`. One record per duplicate group is retained using `pandas.DataFrame.drop_duplicates`.

Missing metadata. Missing values are handled contextually rather than through blanket row removal. Records missing `year` or `month` are excluded from temporal analyses but retained for spatial analyses. Records missing `recordedBy` are assigned the placeholder value `"unknown"` before computation of effort proxies, so that valid observation events without observer attribution are not silently discarded.

4.4 Geographic Scope and Spatial Aggregation

We define Denmark as the geographic scope of the analysis, operationalised using GADM Level 0 and Level 1 shapefiles (version 4.1) [3], which provide the national boundary and the boundaries of Denmark’s five administrative regions respectively. All spatial operations are performed using

`geopandas`. Shapefiles are reprojected to WGS84 (EPSG:4326) to match the coordinate system of the occurrence data.

Strategy A: Regular Grid. We overlay a $0.3^\circ \times 0.3^\circ$ grid on Denmark's bounding box (approximately $25\text{ km} \times 19\text{ km}$ per cell at Denmark's latitude). Each occurrence record is assigned to a grid cell by rounding its coordinates to the nearest grid boundary. The grid is clipped to Denmark's national boundary polygon using `gpd.clip` to remove cells that fall entirely in the ocean or in neighbouring countries.

Strategy B: Administrative Regions. Each occurrence record is assigned to one of Denmark's five administrative regions (Region Hovedstaden, Region Sjælland, Region Syddanmark, Region Midtjylland, Region Nordjylland) via `geopandas.sjoin` with `predicate="within"`. To handle coastal records whose coordinates fall marginally outside polygon boundaries due to GPS rounding, unmatched records are resolved using `gpd.sjoin_nearest`, applied after reprojecting both points and polygons to EPSG:25832 (UTM Zone 32N) to ensure distance calculations are performed in metres rather than degrees. **This is a case of iterative refinement, where we had wrongly calculated distances in degrees.** The spatial join is executed in chunks of 200,000 rows with incremental Parquet output to prevent kernel crashes from accumulated GeoDataFrame memory. We decided against using this and went ahead with national parks instead.

Strategy C: National Parks. The locations of Denmark's five national parks were incorporated into the analysis. Park locations were approximated using central coordinates derived from public geographic sources. Circular buffers were created around these coordinates to represent park areas:

- Thy National Park
- Mols Bjerge National Park
- Wadden Sea National Park
- Skjoldungernes Land National Park
- Kongernes Nordsjælland National Park

4.5 Biodiversity Indicators

Species Richness. For each spatial unit (grid cell), species richness is defined as the number of unique species recorded within that unit. It is computed using `pandas.groupby` on the spatial unit identifier followed by `nunique` on the `species` column.

Range Restriction. For each species, occupancy is defined as the number of distinct spatial units in which it was recorded. This is computed by `groupby species` followed by `nunique` on the spatial unit identifier. Species in the bottom 10th percentile of occupancy are classified as range-restricted, representing taxa whose recorded distributions are geographically concentrated within Denmark and which may be of heightened conservation concern.

Observation Effort. Observation effort was measured using the number of occurrence records per grid cell. Sampling effort is an important factor in biodiversity analysis because regions with more observations may appear artificially more diverse.

Sampling effort was therefore calculated as:

Sampling Effort = Total number of observations per grid cell

By visualizing sampling effort alongside species richness, the analysis identifies areas where biodiversity patterns may be influenced by uneven sampling intensity.

4.6 Hotspot Identification

We define biodiversity hotspots as spatial units whose species richness exceeds 60% of the total number of unique species recorded across the entire dataset. With a total of 510 unique species in the filtered dataset, this corresponds to a threshold of $0.60 \times 510 = 306$, so units with 306 or more unique species are designated hotspots.

4.7 National Park Overlap Analysis

National park boundaries for Denmark's five national parks (Nationalpark Thy, Nationalpark Mols Bjerger, Nationalpark Vadehavet, Nationalpark Skjoldungernes Land, Nationalpark Kongernes Nordsjælland) were obtained from OpenStreetMap via an Overpass Turbo query using Denmark's bounding box. Overlap between hotspot grid cells and national park polygons was assessed using `gpd.overlay(how="intersection")`, which computes the actual geometric intersection between grid cell polygons and park polygons. A grid cell is counted as overlapping a park if any portion of its polygon intersects the park polygon, rather than requiring the cell's centroid to fall inside the park. This is the more conservative and analytically correct criterion, as it captures cells straddling park boundaries.

4.8 Dashboard

- **Data Ingestion and Preprocessing:** The complete GBIF dataset for the Denmark bird biodiversity analysis was initially hosted on Hugging Face. To optimize the initial development phase, a representative subset of the data was extracted. This subset filtered for essential columns and was saved as a lightweight CSV file to prototype the basic dashboard layout.
- **Initial Dashboard Prototype:** The first iteration of the dashboard was built around a foundational interactive bar plot designed to visualize raw species counts.
- **API Integration for Species Metadata:** A Wikipedia API call was integrated to dynamically fetch species summaries and reference images. To account for indexing variations, a robust fallback search logic was implemented: if a query using the scientific name failed to return an image, the system automatically queried the species' common name to retrieve the corresponding text and media.
- **Status Classification System:** A custom filtering script (`filter.py`) was developed to categorize each species based on its ecological and conservation status, applying specific tags such as *endangered*, *exotic*, *migratory*, *common*, *extinct*, or *unknown*.
- **Geospatial Distribution Mapping:** Distribution plots were generated for all 510 species. To manage asset loading efficiently without bloating the application, these plots were hosted on Cloudinary. A custom script (`cloud_image.py`) collected the API endpoints for these images, which were then merged into a central `species_status.csv` file, allowing the dashboard to render distribution maps dynamically based on user selection.
- **Final Analytical Visualizations:** Following the completion of the core analysis, the final static plots were generated and similarly hosted on Cloudinary. The corresponding image URLs were cataloged in a `plots.csv` file and integrated into the dashboard via a dropdown menu, enabling seamless user navigation through the finalized analytical results.

Metric	Count
Total rows processed	40,319,270
Valid rows kept	30,172,537
Filtered out	5,441,626
Malformed (skipped)	4,705,107

Table 2: Record counts from the inline streaming filters stage.

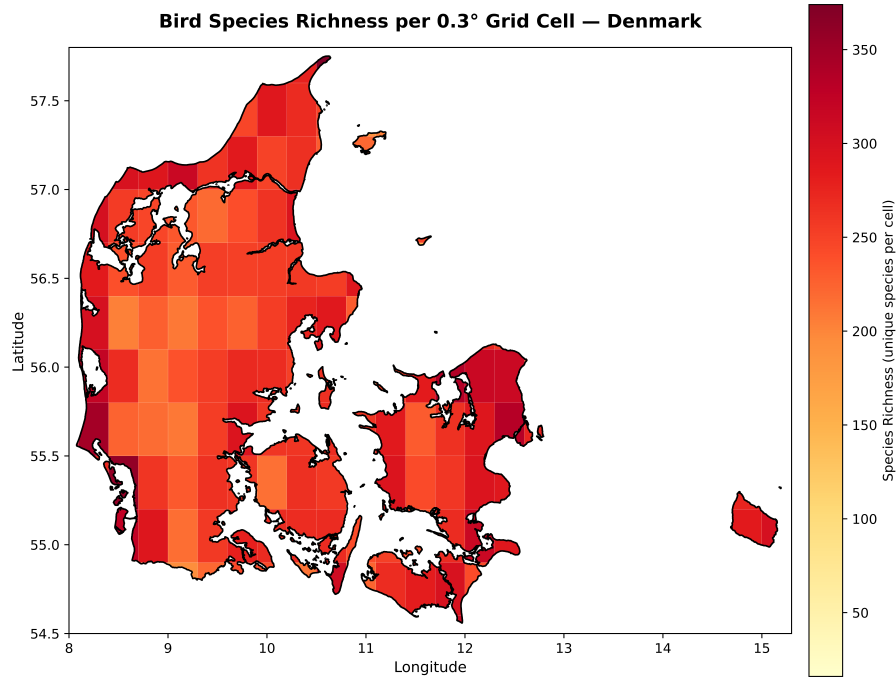


Figure 1: Raw species richness per 0.3° grid cell across Denmark. Grey cells had no occurrence records in the filtered dataset. The concentration of high-richness cells near urban centres and coastlines reflects observer effort density rather than ecological signal.

5 Results - Plots and Analysis

5.1 Ingestion and Cleaning Summary

The streaming pipeline processed all records in the raw dataset. After applying the inline filters (class, coordinate validity, geospatial issues, occurrence status, taxonomic rank, species name, and year ≥ 2000), a subset of the original records was retained as Aves records identified to species level with valid spatial and temporal metadata. Subsequent cleaning removed additional records for taxonomic mismatch flags, coordinate uncertainty above 10 km, and confirmed duplicates. The final analysis dataset contains records spanning species accumulated across decades of volunteer observation year 2000 onwards. Table 2 summarises the record counts.

5.2 Species Richness

Figure 1 shows the raw species richness map under the 0.3° grid aggregation. The highest-richness cells are concentrated along Denmark's coasts and in the vicinity of major cities, particularly in the Copenhagen metropolitan area and around Aarhus. This pattern is consistent with observer effort concentration rather than ecological preference, as these areas have the highest human population density and consequently the most active birdwatchers.

5.3 Temporal Sampling Effort

To examine how data collection has changed over time, the total number of observation events was calculated for each year in the dataset. An observation event was defined as a unique combination of observer identity and observation date.

Figure 2 shows a strong increase in sampling effort from the early 2000s onward. The number of observation events grows steadily until approximately 2020, indicating increasing participation in biodiversity monitoring and citizen science programs. After 2020, sampling effort remains relatively high but fluctuates slightly between years.

The sharp decrease observed in the most recent year (as 2026 has just started) likely reflects incomplete reporting or partial data coverage rather than a true decline in observation activity.

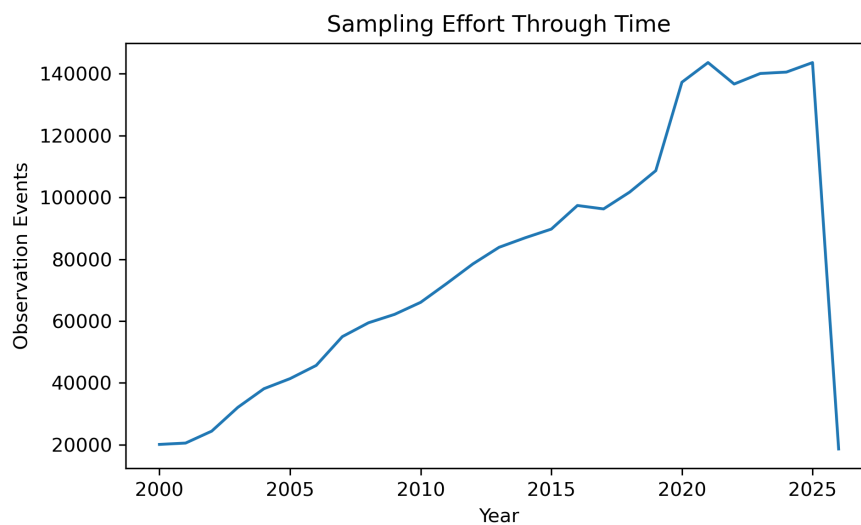


Figure 2: Temporal trend in bird observation events across Denmark. Sampling effort has increased substantially over the past two decades, reflecting growing participation in biodiversity monitoring.

5.4 Spatial Sampling Effort

Sampling effort was also analysed spatially by counting the number of unique observation events within each grid cell.

Figure 3 shows the geographic distribution of sampling effort across Denmark. The results reveal substantial spatial heterogeneity in observation intensity. Certain regions, particularly in eastern Denmark, exhibit very high sampling effort, while large areas in western and northern Denmark show comparatively lower observation densities.

These differences likely reflect variations in human population density, accessibility, and proximity to urban centers or popular birdwatching locations.

5.5 Relationship Between Sampling Effort and Species Richness

To assess whether biodiversity patterns are influenced by observation intensity, species richness was compared directly with sampling effort at the grid-cell level.

Figure 4 shows a strong positive relationship between observation effort and observed species richness. Grid cells with greater sampling effort tend to report higher numbers of species. This relationship indicates that part of the observed biodiversity variation may result from differences in sampling intensity rather than purely ecological factors.

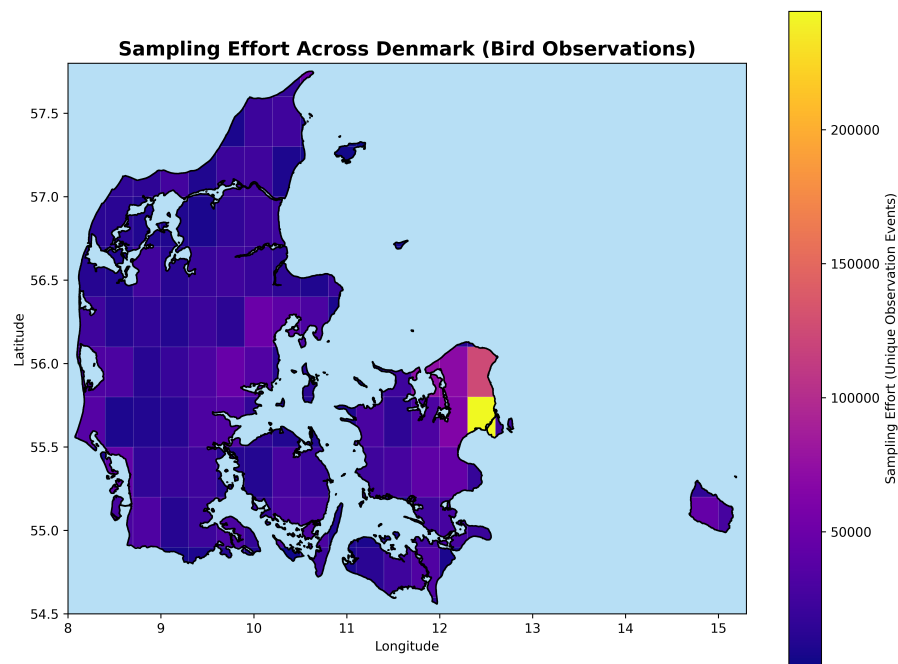


Figure 3: Spatial distribution of sampling effort across Denmark. Grid cells with higher numbers of observation events appear in brighter colours. Substantial spatial variation in observation intensity is visible across the country.

The logarithmic scaling of sampling effort highlights this relationship across multiple orders of magnitude, confirming that heavily sampled areas consistently report higher biodiversity counts.

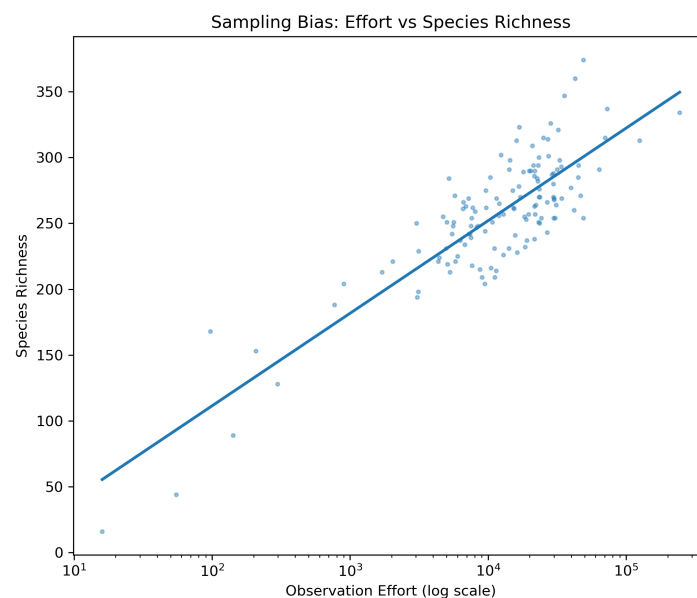


Figure 4: Relationship between sampling effort and observed species richness. Each point represents a spatial grid cell. The positive trend indicates that higher observation effort is associated with higher recorded biodiversity.

5.6 Implications for Biodiversity Analysis

These results demonstrate that sampling effort plays a significant role in shaping observed biodiversity patterns. Areas with higher observation intensity are more likely to report larger numbers of species simply because they are sampled more frequently.

Therefore, interpretation of biodiversity hotspots must consider potential sampling bias. While some regions identified as biodiversity hotspots likely represent genuine areas of high ecological diversity, others may partially reflect concentrated observation activity.

Accounting for sampling effort is therefore essential for robust biodiversity assessments and for identifying true conservation priorities.

5.7 Hotspots and Overlap with National Parks

Figure 5 shows the hotspot map with national park boundaries overlaid. Hotspot cells (those with ≥ 306 unique species, representing $\geq 60\%$ of the 510-species total) are shown in red, with national park boundaries shown as dashed green outlines.

5.7.1 Implications for Hotspot Identification

The comparison demonstrates that spatial aggregation strategy can significantly influence the interpretation of biodiversity patterns.

Grid-based aggregation provides a high-resolution view of biodiversity distribution, allowing localized hotspots to be detected. In contrast, administrative regions such as national parks aggregate biodiversity over larger spatial units and may mask fine-scale patterns.

As a result, some biodiversity hotspots identified using grid cells occur outside national park boundaries. These areas may represent potential conservation gaps where high biodiversity exists but formal protection is limited.

Overall, combining grid-based biodiversity analysis with administrative spatial units provides a more comprehensive understanding of biodiversity distribution and can help inform conservation planning by identifying both protected and unprotected areas of ecological importance.

5.7.2 High Biodiversity Regions with Low Observation Effort

To identify potentially under-sampled biodiversity regions, species richness and observation effort were compared across spatial grid cells. Species richness was calculated as the number of unique species recorded within each grid cell, while observation effort was defined as the number of unique observation events.

Grid cells were classified as potential under-sampled biodiversity regions if they satisfied two criteria: relatively high species richness and comparatively low observation effort. Thresholds were defined using quantiles of the dataset, where grid cells above the 60th percentile of species richness and below the 40th percentile of observation effort were selected.

The analysis identified five grid cells that satisfied these conditions (Table 3). These cells exhibited species richness values between 269 and 285 species despite moderate observation effort levels ranging from approximately 5,200 to 10,400 observation events.

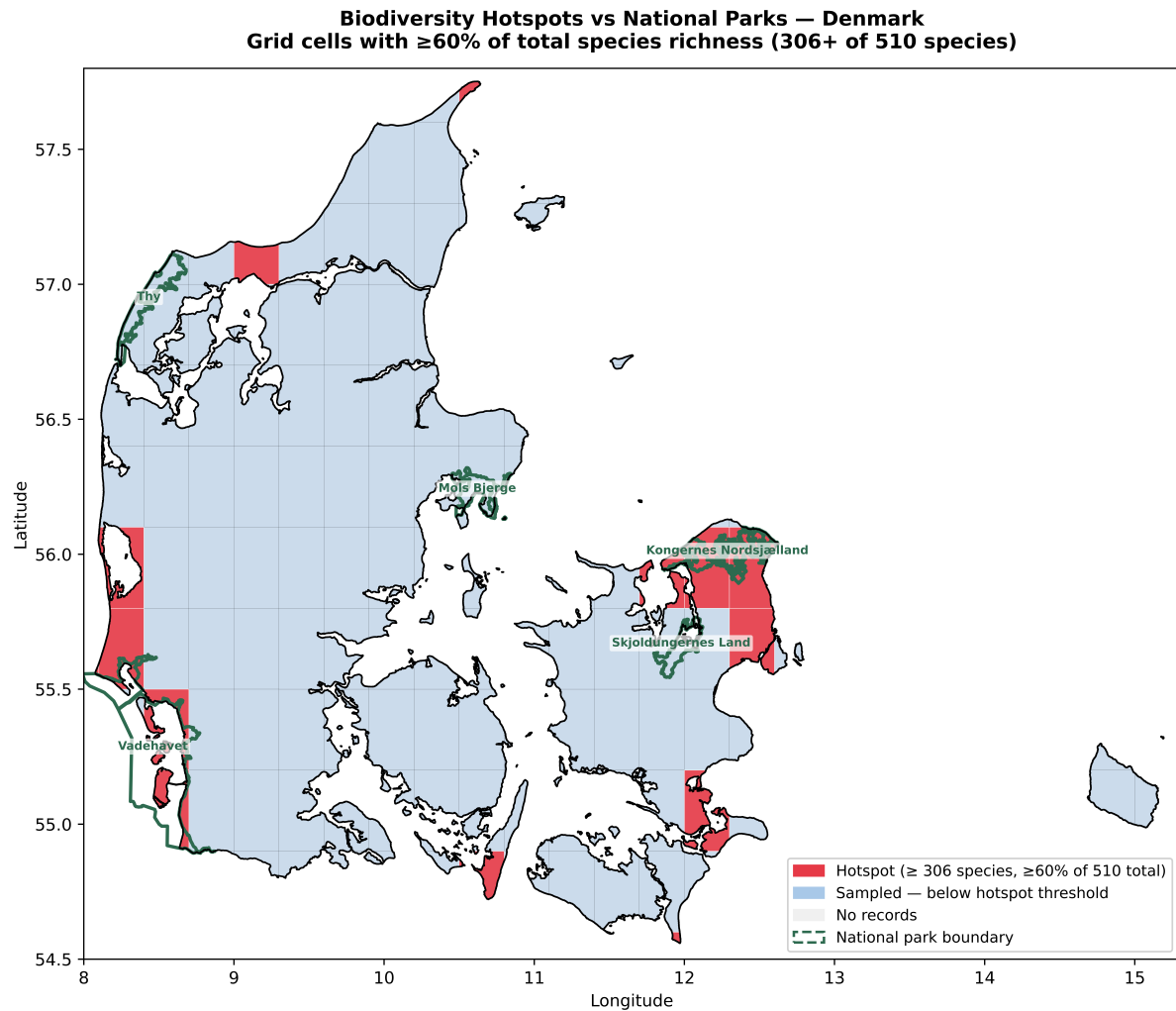


Figure 5: Biodiversity hotspot cells (red) overlaid with Denmark's five national park boundaries (dashed green). Hotspot cells are defined as those with species richness $\geq 60\%$ of the total (306+ of 510 species). The overlap analysis reveals both alignments between protected areas and biodiversity peaks, and gaps where hotspot cells fall outside any national park.

Table 3: Grid cells with high biodiversity but relatively low observation effort.

Grid Cell ID	Species Richness	Observation Effort	Coordinates (Lat, Lon)
lat_55.2_lon_15.0	284	5209	(55.2, 15.0)
lat_55.2_lon_8.1	271	5726	(55.2, 8.1)
lat_55.5_lon_10.5	269	7222	(55.5, 10.5)
lat_56.1_lon_12.0	275	9611	(56.1, 12.0)
lat_56.4_lon_8.1	285	10377	(56.4, 8.1)

These locations represent areas where relatively high biodiversity has been observed despite limited observation activity compared with heavily sampled regions. Such areas may therefore represent under-sampled biodiversity hotspots.

5.7.3 Spatial Interpretation

The identified grid cells are distributed across multiple regions of Denmark, including both western and eastern parts of the country. The presence of high species richness in these areas suggests that they may contain ecologically diverse habitats such as coastal environments, wetlands, or migratory bird corridors.

Because these areas have received comparatively fewer observation events than other regions, the true biodiversity present may be even higher than currently documented.

5.7.4 Implications for Conservation and Monitoring

Identifying regions with high biodiversity but relatively low observation effort has important implications for biodiversity monitoring and conservation planning.

First, these areas represent priority targets for future biodiversity surveys. Increasing observation effort in these locations could reveal additional species records and improve understanding of regional biodiversity patterns.

Second, these locations may represent conservation-relevant habitats that are currently underrepresented in biodiversity datasets. If future surveys confirm their ecological importance, they could be considered for enhanced monitoring or conservation initiatives.

Finally, this analysis highlights the importance of accounting for sampling bias when interpreting biodiversity patterns. Without considering observation effort, areas with intensive sampling may appear artificially more biodiverse than regions that have simply been less surveyed.

By integrating both biodiversity metrics and observation effort, spatial analyses can more effectively identify locations where additional data collection would yield the greatest ecological insight.

5.8 Iterative Refinement of the Analysis

During the development of the biodiversity analysis pipeline, several iterative refinements were made to improve the reliability and interpretability of the results. These refinements involved adjustments to data cleaning procedures, spatial aggregation strategies, and visualization techniques based on exploratory analysis findings.

5.8.1 RAM heavy operations

For computationally intensive operations such as spatial joins and biodiversity metric calculations, we initially attempted to perform these operations on the entire dataset at once. However, this approach led to kernel crashes due to excessive memory usage. To address this issue, we implemented a chunking strategy where the dataset is processed in smaller batches (e.g., 200,000

rows at a time) with incremental output to disk. This allowed us to complete the analysis without exceeding memory limits.

5.8.2 Refinement of Spatial Aggregation

For the grid-based spatial aggregation approach, we had initially created grid cells with 0.1° resolution. Since this resulted in too many cells, we decided to reduce the resolution to 0.3° .

5.9 Data Limitations, Assumptions, and Potential Sources of Bias

While the dataset used in this study provides extensive biodiversity observations across Denmark, several limitations and potential sources of bias must be considered when interpreting the results.

5.9.1 Sampling Bias

Exploratory analysis revealed a strong relationship between sampling effort and species richness, indicating that areas with more observation events tend to report higher numbers of species. This pattern suggests that some biodiversity hotspots identified in the analysis may partially reflect observer activity rather than purely ecological processes.

5.9.2 Spatial Resolution Assumptions

Different spatial aggregation strategies may therefore produce slightly different hotspot patterns. For example, biodiversity patterns interpreted using grid cells may differ from patterns observed using administrative boundaries such as national parks.

5.9.3 Temporal Limitations

The dataset contains observations spanning multiple years, and observation effort has increased substantially over time due to expanding citizen science initiatives and biodiversity monitoring programs. Consequently, more recent years may contribute disproportionately to observed biodiversity patterns.

This temporal variation in observation effort may influence spatial biodiversity estimates, particularly in regions that have experienced rapid growth in observation activity.

5.9.4 Observer and Identification Bias

Species occurrence datasets often rely on contributions from multiple observers with varying levels of expertise. Differences in species identification skills, reporting practices, and survey focus can influence which species are recorded in different locations.

For example, conspicuous or easily identifiable bird species may be reported more frequently than rare or difficult-to-identify species. Such biases can affect estimates of species richness and spatial biodiversity patterns.

6 Results - Dashboard

The interactive dashboard is accessible by running the Dash application `dashplot.py` locally from the project repository. The species distribution panel displays individual occurrence records for any of the 510 species as geo-referenced dots on a map of Denmark, loaded from pre-generated static plots.

There is an option to filter among the species list by biological characteristics.

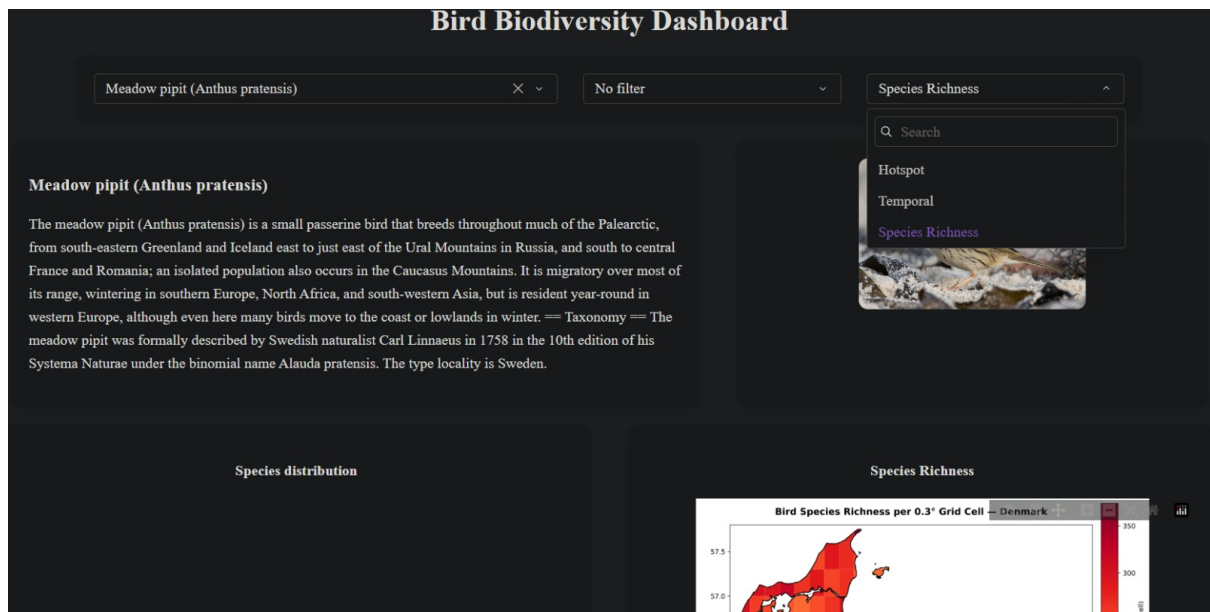


Figure 6: A screenshot of the dashboard.

Another section displays the richness, observation effort, and hotspot maps, accessible via a dropdown menu.

Figure 6 shows a static image of how the dashboard looks like.

7 Results - Use case

The identified hotspot regions can be used for conservation prioritization. Regions with high species richness may represent important ecological communities that require protection.

By identifying these areas quantitatively, conservation planners can better allocate resources and focus efforts on regions that support the highest levels of biodiversity.

8 Conclusion and Outlook

In this study, we analyzed bird species richness across Denmark using a spatial grid-based approach. By calculating the number of unique bird species in each grid cell, we were able to map spatial patterns of bird biodiversity.

Using a hotspot threshold of 306 species per grid cell, we identified 14 biodiversity hotspot regions across Denmark. Several grid cells showed exceptionally high richness, with more than 330 recorded species.

These hotspots may correspond to ecologically important habitats such as coastal areas, wetlands, and migratory bird stopover locations.

Future work could extend this analysis by incorporating environmental covariates (land cover, elevation, distance to coast, etc.) which would allow ecological correlates of richness to be disentangled from sampling effort.

9 Acknowledgements

We acknowledge DOF/BirdLife Denmark for making the DOFbasen database publicly available via GBIF, and the Global Biodiversity Information Facility for providing open access to biodiversity data infrastructure. We also acknowledge the OpenStreetMap contributors whose data, accessed via Overpass Turbo, provided the national park boundary polygons used in the overlap analysis. Parts of this project were developed with assistance from Claude (Anthropic, 2025).

10 Author Contributions

Adhin, Sumit, Aritra, L were responsible for the data ingestion pipeline, including the streaming script, batch Parquet writing, and initial quality filtering.

Aritra was responsible for data cleaning, geographic scope definition, and creating .gpkg files for the two spatial aggregation strategies including the grid, national parks, and regions.

L was responsible for all the biological analysis parts, including biodiversity indicator computation, and the hotspot and national park overlap analyses.

L and **Aritra** were responsible for the visualizations of the biodiversity patterns, including the species richness maps, sampling effort maps, and hotspot maps.

Adhin and **Sumit** were responsible for making the entire the interactive Dash dashboard, and visualization of species distribution maps.

11 Code availability

The code for this project is available at the following GitHub repository: https://github.com/Lbhagirathi/Biodiversity_dataset_analysis-.git. The repository includes all analysis notebooks, the streaming ingestion script, the dashboard application, and a `README.md` with setup instructions. The raw dataset is not committed to the repository due to file size; it is accessible via the GBIF DOI [2] and the HuggingFace mirror documented in the `data/README.md` file.

AI assistance (Claude, Anthropic, 2025) was used in developing the streaming ingestion pipeline, the chunked spatial join, and portions of the dashboard code.

12 References

References

- [1] Global Biodiversity Information Facility. GBIF: Free and open access to biodiversity data, 2024. URL <https://www.gbif.org>. Accessed: 2026.
- [2] T. Nyegaard. DOF/BirdLife Denmark - observations from DOFbasen, 2026. URL <https://doi.org/10.15468/tww7cj>. Occurrence dataset accessed via GBIF.org on 2026-03-09.
- [3] University of California, Davis. GADM: Database of global administrative areas, version 4.1, 2022. URL <https://gadm.org>. Accessed: 2026.

13 Appendix

13.1 Additional Figures

All supplementary plots/figures are available in the `outputs/` directory of the project repository.